

Langerhans Cell Histiocytosis

AT A GLANCE

A group of rare disorders that are considered reactive but have a broad spectrum of severity.

The reactive versus neoplastic nature of LCH cells is debated. Although clonality has been demonstrated in all forms of LCH, this does not necessarily indicate malignant behavior. A clonal origin in genetically predisposed individuals is the most probable hypothesis.

Histopathologic features consist of a dense infiltrate of histiocytes with strong epidermotropism. The unifying element is the typical "LCH cell"—a histiocyte with an irregular, vesiculated, often reniform nucleus and abundant, slightly eosinophilic cytoplasm. These cells are positive for S100 protein, CD1a, and CD207 (langerin), and contain cytoplasmic Birbeck granules.

Pathogenesis

The nature of LCH—reactive versus neoplastic—remains controversial. Evidence supporting a **reactive hypothesis** includes:

1. LCH cells are highly differentiated.
2. Granulomatous lesions resemble those seen in infections or foreign-body reactions.
3. Lesions may spontaneously regress.

Proposed pathogenic steps in the reactive model:

1. Viral infection induces activation of histiocytes.
2. Immunologic mechanisms drive the pathologic Langerhans cell phenotype.
3. A cytokine-mediated inflammatory storm leads to lesion formation.

Numerous viruses—such as human herpesvirus 6, cytomegalovirus, adenovirus, and parvovirus—have been implicated in disease initiation, but none has been definitively proven as causative.

In contrast, evidence supporting a **clonal neoplastic origin** includes:

1. Bone marrow origin and clonality of LCH cells.
2. Rare familial cases of LCH.
3. Chromosomal instability, including abnormalities on chromosome 7 and p53 protein overexpression.
4. Co-occurrence of LCH and myelodysplastic syndrome.

Two explanations have been proposed for the association with myelodysplasia:

- Marrow dysplasia may be secondary to cytokine-driven infiltration by LCH cells, rather than a true myelodysplastic syndrome.
- Alternatively, both LCH and myelodysplastic cells may arise from a common clonal pluripotent dendritic (stem) cell, suggesting LCH could be classified as a "myeloid dendritic stem cell disorder."

These findings imply a genetic predisposition and have prompted research into candidate genes involved in LCH pathogenesis.

Clinical Findings

Classification

Langerhans cell histiocytosis (LCH) presents across a wide clinical spectrum—from localized, asymptomatic disease to systemic, aggressive multisystem involvement. Clinical manifestations depend on the extent and location of lesions.

Historically, LCH has been categorized under several names based on clinical presentation:

- **Letterer-Siwe disease (LSD):** Acute, disseminated, multisystem form typically affecting infants and newborns; untreated cases are often fatal.
- **Hand-Schüller-Christian disease (HSC):** Chronic, progressive, multifocal form usually presenting in childhood.
- **Eosinophilic granuloma (EG):** Localized, benign form, often limited to a single bone lesion.

- **Hashimoto-Pritzker disease (HPD):** Self-healing, congenital variant presenting at birth or within the first days of life.

The Histiocyte Society has established a standardized classification system for histiocytoses (refer to Table 148-1).